

2.2b

Thinking, Problem Solving, Judgements, & Decision Making

Solving Problems & Making Decisions

We often have sub-par executive functions i.e. high-level cognitive activities for planning, organising & reasoning.

How do we solve problems?

Some problems we solve with algorithms i.e. step-by-step procedures guaranteed to give a solution.

But solving some problems algorithmically can be laborious & unnecessarily time consuming.

So we also use Heuristics, simpler thinking strategies. These mental shortcuts are speedier, but more error-prone than algorithms.

Sometimes we don't use any strategy, arriving at a solution through insight.

Brain scans show bursts of activity in RTL at the instant of discovery.

Our mental Stubbornness:

Confirmation Bias leads us to seek evidence that reinforces our own beliefs.

Once we get hung up on an incorrect view of a problem, it's hard to approach it from another angle
∴ fixation

Mental Set is an ex. of fixation, it's our tendency to approach a problem with a 'mindset' of what has worked previously

What is Intuition?

We rarely take time & effort to systematically reason each judgement & decision

Instead we follow our intuition: our fast, automatic, unreasoned feelings & thoughts.

Representative heuristic is an ex. of intuition: judging likelihood by comparing w/ prototypes

People may unconsciously use the representativeness heuristic with their racist intuition, creating racial bias.

If people observe random events happening repeatedly, they may use the rep. heuristic to judge likelihood of future events $\therefore$ gambler's fallacy.

Another ex. of int. is the availability heuristic. We evaluate commonality of an event based on its mental availability.

The av. heuristic means that very vivid and graphic info seems very commonplace.

How overconfident are we?

We tend to overestimate the accuracy of our knowledge and judgement $\therefore$ overconfidence.

Overconfidence causes planning fallacy $\therefore$ overestimating my future leisure time & income

Overconfidence also fuels sunk-cost fallacy $\therefore$ we stick to our original plan because we've invested time even if switching could save time

Belief Perseverence:

Our belief perseverance is our tendency to cling to our beliefs in the face of contrary evidence.

Studies where participants were given 2 studies, one agreeing w/ them & 1 disagreeing. People were more impressed w/ the study agreeing w/ them & disputed the other.

Rather than use evidence to draw conclusions, they used conclusions to assess evidence: motivated reasoning.

Framing:

Framing :: the way we present an issue, can be a powerful tool of persuasion.

Framing solutions as 'gains' is much more popular than as 'losses'.

Gentle framing can nudge people into making beneficial decisions.

How Can Intuition be Powerful?

Int. is recognition born of exp.; implicit (unconscious) knowledge

Int. is usually adaptive. We rely on learned associations that surface as gut feelings.

Int. is huge. Thru selective attention, we focus on a fraction of stimuli. Our subconscious uses everything else. Unconscious, automatic influences constantly affect our judgement.

Letting a problem incubate while we attend to other things or especially while asleep can solve it.

Personal Connection

MCQ:

There have been many times where a coding problem I couldn't figure out was solved subconsciously in my sleep. This has happened SO MANY times that stopping and returning has become a standard part of my process.

1. b 2. a 3. c 4. c

5. d 6. c 7. c 8. a

9. b